Non-destructive investigation of samples by 'Magnetoscopy'

Introduction : Magnetoscopic testing (MT) is non-destructive testing that reveals defects on the surface or slightly below it. These surface defects deform the magnetic field created by a device, causing leakage paths visible to the naked eye after the application of a fine iron powder. This inspection method applies to ferromagnetic materials only.

Description of the products and material necessary :

Cleaning/washing products : SKC-S/acetone

Penetrating product : 7HF or Fluxo 3– ferromagnetic material black in color (final observation conducted in daylight)

14HF or 410 HF (fluorescent product observation under UV lamp)

Contrasting product : WCP2 or Fluxo 7

Excitation of fluorescent product : UV-Lamp

5 samples made of ferromagnetic materials are provided. You have to study any 2 samples out of the 5 samples as per your choice. The samples are numbered on the flat surface. You can select one of the following number : 1, 2, 4, 9, 21

Some important information before starting the experiment:

Wear your lab-coat and gloves before starting the experiment

Time required for the product to penetrate varies between 5 to 10 mins.

The samples can be washed with water with a final cleaning consisting of a tissue dipped in SKC-S (if necessary)

The contrasting product is applied very delicately to make an extremely thin layer.

The minimum delay necessary before observation is 3 mins.

Switch on the UV-lamp before you start the expts. It requires time to give the max. output intensity.

Procedure :

- **Wash the samples with water (or with acetone in case they were dirty). Dry them properly with a tissue paper. On one sample you spray Fluxo 3. You coat all the faces of the sample with the product. During the spraying step, hold it in the wash basin.**

- **On the other sample you spray 410HF. You coat all the faces of the sample with the product. Note down on which sample you have sprayed which product.**

- **Place the sample vertically in the basin and allow the sample to rest for 5 or 10 mins without disturbing it.**

- **Next DRY the sample delicately with a tissue paper. DO NOT WASH THE SAMPLE.**

- **Then extremely delicately spray the samples with the contrasting product. For spraying delicately keep a large distance between your sample and the spray.**

- **Allow the contrastant to penetrate inside. For doing this keep the sample vertically in the basin for a time period of 5 mins.**

- **Next, magnetise the two samples using the electromagnet Y6. Magnetisation is conducted by placing the two poles of the electromagnet on the sample to be magnetized and at the same time pressing the switch located below the electromagnet during a time period of 15 to 20 secs. Check whether the sample has got magnetized with the help of the magnetic needle (compass).**

- **Observe the sample for defects using daylight for sample sprayed with Fluxo 3 and uv- lamp for sample sprayed with 410HF.**

- **Analyse the defects. You can take photos of the sample and put them in your expt. report.**

- **Finally, wash and dry the sample. Next step is to demagnetize it. For demagnetization, place the electromagnet on the sample. With one hand hold the sample firmly down on the table. Press the switch located below the electromagnet with the other hand and while pressing, at the same time try to pull the electromagnet away from the sample. In this manner the sample is demagnetized.**

- **Arrange the sample in the box.**

Q1. For every sample determine the type of defect and its position. You can either draw a schematic or put a photo of the sample pointing out the defects with the help of arrows

Q3. Determine in the two cases which method is better and why ?

Q4. Comment on the different type of defects which you observed and explain them in brief.